Day 5 — Performance Baselines, Monitoring & Validating AI-Generated Data Summaries

> **Activity packet** for participant quads. Today's job: write **Section 5** of the TMD — performance baselines, monitoring plan, and AI-summary validation log. Then **assemble the full TMD**, peer-review it, and ship it as the sibling artifact to TCD and PRD.

Where we are in the week

The TMD has Sections 1–4 (data, cloud, API, sequences). Today produces Section 5 — **performance baselines + monitoring plan + AI-summary validation log** — and the integrated TMD ships.

The AI-summary validation log is the **new discipline this week**. AI is increasingly used to summarize dashboards, query results, and metric reports. The TPM job: validate those summaries against the source data and document the result.

Today's cadence

Same as Mon–Thu (no special split like Week 3 Day 5 had). Block 4 is the cross-review + sign-off.

Inputs

- TMD Sections 1–4 (Days 1–4)
- TCD Section 4 SLOs (anchor for baselines + targets)
- The Tier Sheet from Week 2 — operational signals point at what to monitor
- The PRD — context for what dashboards stakeholders need

What Section 5 contains

```
## 5. Performance baselines & monitoring

### Baselines
- Pre-launch numbers we measure today (or expect at
launch)
- Historical context where applicable

### Targets
- From TCD Section 4 SLOs (do not redefine —
reference)

### Monitoring plan
- Dashboards (operator-level, executive-level)
- Alerts (thresholds, severities, on-call routing)
- The "fire before users notice" check

### AI-summary validation log
- Each AI-assisted summary used in this TMD or
referenced in
  upstream artifacts: prompt, output, validation
status
```

Performance baselines — measuring "where we start"

A baseline is what the metric reads **today** (or at launch, for new features). Without a baseline, you can't tell if you've improved.

Two kinds of baselines

Kind	When	Source
Existing-system baseline	Brownfield work — improving something that exists	Production data (current dashboards, query histories)
Expected-launch baseline	Greenfield work — new feature	Estimated from comparable features, load-test data, or "we expect X based on [reasoning]"

For greenfield, the discipline is to **state the assumption** rather than skip the baseline. "We expect ~200 reconcile-submits/day in the first 30 days, based on dispatcher count × shifts/day × (estimated adoption rate)." That assumption gets tested at launch + 7 days.

What to baseline

For each TCD Section 4 SLO, capture the corresponding baseline:

- **Latency:** what the system does today (or expects to)
- **Availability:** what the system has done historically
- **Throughput:** current volume + projected
- **Error rate:** historical or expected

Plus any **operational signal** from the Tier Sheet that doesn't show up in the SLOs.

The monitoring plan — fire before users notice

A monitoring plan has three layers:

Layer	Audience	Rhythm
On-call alerts	Engineers on rotation	Real-time
Operator dashboards	TPM, eng lead, SREs	Daily / weekly
Executive dashboards	Leadership	Weekly / monthly

Each layer has a different question:

- On-call: "Is something on fire right now?"
- Operator: "Is the system trending the right way?"
- Executive: "Are we meeting our SLOs / KPIs / NS?"

Alert design — the four-question check

For each alert, answer:

1. **What does it mean?** (Plain English)
2. **What does the on-call do when it fires?** (Specific runbook step)

3. **What's the threshold + window?** (e.g., "5xx rate > 1% over 5 min")
4. **What's the severity?** (Page / Slack / log only)

If you can't answer all four, the alert is noise — it'll be muted within a sprint.

AI-summary validation — the new discipline

This week's prompts produced AI-assisted critiques, summaries, and structuring. Today's discipline: **validate each AI-assisted summary** against the underlying data and document the result.

Why this matters

AI summarization of data — query results, dashboards, metric reports, performance test outputs — is increasingly common. The failure modes:

Failure	What happens
Wrong number	AI cites "p95 = 800ms" when the chart shows 1200ms
Wrong direction	AI says "improving" when the trend is flat or declining
Wrong attribution	AI explains a spike with the wrong cause
Missing context	AI summarizes the headline; misses the relevant footnote

The validation procedure

For each AI-assisted summary used in any week's artifact:

```
| # | Summary purpose | Prompt (link) | Source data | Validation |
|---|-----------------|---------------|-------------|------------|
| 1 | Strategy brief pain themes (Wk 2 D4) | Pattern Lib #3 | 14 tickets + 3 interviews | Cross-checked all citations; 1 hallucinated; cut |
| 2 | Architecture critique (Wk 4 D1) | Pattern Lib #5 | TCD Section 1 stance | Adopted 2/3 objections; rejected 1 (irrelevant scenario) |
| 3 | API contract critique (Wk 5 D3) | Pattern Lib #8 | TMD Section 3 | Adopted 4/5 issues; deferred 1 to v2 |
| 4 | Cloud topology risk scan (Wk 5 D2) | Pattern Lib #7 | TMD Section 2 | All 3 valid; 1 added as TCD risk; 2 already known |
```

The log is **cumulative across weeks** — the team should know which AI outputs have been validated and which are pending.

Status values

- **Cross-checked all citations** (best — every claim verified against source)
- **Spot-checked** (rougher — N claims sampled and verified)
- **Adopted with rationale** (we used the AI suggestion; the suggestion's correctness was self-evident or cheap to verify)
- **Pending** (used as input but not yet validated; track to closure)
- **Rejected** (output was wrong; cut)

Activity 1 — Performance Baselines

Format: Quad • **45 min** • Block 1

Purpose

Set baselines for each SLO + key operational signal.

Setup

Each quad needs the TCD Section 4 SLOs, the Week-2 Tier Sheet, and the baseline template. AI optional.

Quad protocol

1. **List the metrics that need baselines** (5 min). One per SLO + 2–3 op-signals from Tier Sheet.
2. **For each, classify** (5 min). Existing-system baseline or expected-launch?
3. **Capture the baseline value** (25 min). For existing: production number (real or assumed). For expected: the assumption + reasoning.
4. **Set the "test at launch + N days" plan** (10 min). When will we check the baseline against reality?

What "good" looks like

- Every SLO has a paired baseline
- Greenfield baselines name **the assumption** (not "we don't know")
- The "test at launch + N days" plan has dates
- The baseline format is reusable for future TMDs

Deliverable

A baselines table covering every TCD Section 4 SLO and 2–3 Tier Sheet operational signals, each with a value, source/assumption, and a verify-at date.

Worked example — FieldPulse reconcile

```
| Metric | Baseline | Source / assumption | Verify
at |
|--------|----------|----------------------|--------
---|
| Reconcile-flow p95 latency | Expected ~700ms |
Sequence diagram totals + 30% buffer | Launch + 7d |
| Reconcile-flow availability | Expected 99.5% (=
TCD Section 4 target) | Inherits monolith
availability | Launch + 30d |
| Reconcile-submits/day | Expected 200 | Dispatcher
count × 1 reconcile per shift | Launch + 7d |
| Submission error rate | Expected < 2% | Comparable
features in this app run < 2% | Launch + 14d |
| 5xx rate | Expected < 0.5% | Inherits monolith
baseline | Launch + 7d |
```

Activity 2 — The Monitoring Plan

Format: Quad • **50 min** • Block 2

Purpose

Design dashboards + alerts across the three layers.

Setup

Each quad needs the baselines from Activity 1, the four-question alert template, and the three-layer dashboard prompts. AI optional.

Quad protocol

Step 1 — On-call alerts (15 min)

For each alert, fill the four-question template:

```
### Alert: <name>
**What it means:** <plain English>
**What on-call does:** <specific first step>
**Threshold + window:** <X over Y minutes>
**Severity:** Page / Slack / log
```

Aim for **3–5 alerts** per feature. More = noise.

Step 2 — Operator dashboard (15 min)

What does a TPM look at every Monday morning to see if the feature is healthy?

A short list:

- Top-line metric (the SLO you care most about)
- Operational signals trending
- Alert volume / mute reasons
- Any deferrals from the previous week

Step 3 — Executive dashboard (10 min)

What does leadership see in the weekly review?

A shorter list:

- The NS-relevant operational signal (from Tier Sheet)
- The KPI the feature is meant to move
- Status: green / yellow / red against SLOs

What "good" looks like

- Alerts pass the four-question test
- The operator dashboard has < 6 charts (more = nobody reads)
- The executive dashboard has < 4 charts
- "Mute reasons" tracking is included on the operator level — captures the alert-tuning conversation

Deliverable

3–5 alerts answering the four-question template, plus operator and executive dashboard charts lists capped at 6 and 4 respectively.

Activity 3 — AI-Summary Validation Log

Format: Quad • **50 min** • Block 3

Purpose

Walk back through every AI-assisted summary used across Weeks 1–5 and document its validation status.

Setup

Each quad needs the cumulative AI provenance notes from Weeks 2, 4, and 5, the Pattern Library reference, and the validation log template.

Quad protocol

1. **Inventory the AI uses** (25 min). List every AI-assisted output you used in any artifact: prompts from Weeks 2, 4, 5.
2. **For each, capture the validation row** (15 min):
   - Summary purpose
   - Prompt link (Pattern Library entry)
   - Source data the AI was working with
   - Validation status (using the values above)
3. **Highlight any "Pending"** (5 min). For each, decide: validate now, defer to specific date, or accept the risk.
4. **Sanity check** (5 min). Are there places you used AI without logging it? Add them.

Worked log entry

```
| # | Summary purpose | Prompt | Source | Validation |
|---|-----------------|--------|--------|------------|
| 5 | Sequence diagram critique (Wk 5 D4) | Pattern Lib #11 | TMD Section 4 | Adopted 2/3 issues; AI flagged missing audit-publish on weird path — added |
| 6 | API critique (Wk 5 D3) | Pattern Lib #8 | TMD Section 3 | Spot-checked: AI pointed to inconsistent error format; verified vs Section 3; fixed |
| 7 | Cloud risk scan (Wk 5 D2) | Pattern Lib #7 | TMD Section 2 | All 3 valid; 1 added to TCD Section 5 trade-offs; 2 already in stakeholder matrix |
```

What "good" looks like

- Every AI use across Weeks 2–5 has a row
- Status values are honest
- Pending entries have a deadline
- The log is **a tool the next TPM can audit**, not a checkbox

Deliverable

A cumulative AI-summary validation log covering Weeks 2–5, with honest status values and named deadlines for Pending entries.

Activity 4 — TMD Integration + Cross-Review + Sign-Off

Format: Quad • **55 min** + Wrap • Block 4

Purpose

Assemble the full TMD, cross-review with another quad, and sign off. Same Week-3 / Week-4 review pattern.

Setup

Instructor confirms pairings. Each quad needs the full TMD Sections 1–5 draft and the Friday TMD rubric.

Quad protocol

1. **Solo read-through** (10 min). Each member reads Sections 1–5 and marks coherence issues.
2. **Pool issues** (5 min). De-dupe.
3. **Cross-review** (30 min). Pair with another quad. Reviewer scores with the **Friday TMD rubric**:

Dimension	Weight	What "exemplary" looks like
Data model clarity	20%	Entities, keys, indexes, trade-offs all named
Cloud topology realism	15%	Region/AZ stance fits SLOs; tenancy honest
API contract sharpness	20%	Resources noun-shaped; idempotency, errors, versioning explicit
Sequence model completeness	20%	Happy + sad/weird; protocols + invariants
Baselines + monitoring	15%	Baselines real; alerts pass 4-question test
AI-validation discipline	10%	Log distinguishes correct / partly correct / wrong with evidence

4. **Adopt / defer / reject** (10 min). Same Week-3 discipline.
5. **Sign-off** (no separate block — done within Activity 4). Status: "Approved" or "Approved with gaps."

Deliverable

Signed-off TMD (Approved or Approved with gaps) with adopted cross-review findings, final rubric score, and cumulative AI-validation log attached.

Wrap (last 15 min)

Each quad shares:

- The **single sharpest insight** the TMD added beyond the TCD
- The **one open question** for Week 6 stakeholder negotiation
- Their **TMD final score** (peer + instructor average)

End-of-week (Week 5) checkpoint

Each quad ships:

- [x] **Full TMD** with Sections 1–5 integrated
- [x] Section 1 Data model
- [x] Section 2 Cloud topology + ROM cost
- [x] Section 3 API contract with idempotency / versioning / errors
- [x] Section 4 Three sequence diagrams (happy / sad / weird)
- [x] Section 5 Performance baselines + monitoring plan + AI-validation log
- [x] Cross-reviewed + signed off
- [x] Cumulative AI-validation log across Weeks 1–5

The TMD goes into Week 6 alongside the PRD and TCD. Week 6 (Stakeholder Alignment & Negotiation) **negotiates** the constraints documented across all three artifacts.